

MATH 8090: Model-Free Estimation of Stationary Means and Covariances

Readings: CC08 Chapter 4.3, 3.2, 3.6; BD16 Chapter 2.4; SS17
Chapter 1.5

Whitney Huang



Week 4

Agenda

The Autoregressive Process

Model-Free Estimation of Stationary Means and Covariances

Testing Temporal Dependence

The Autoregressive Operator

- ▶ AR(1) process

$$\begin{aligned}\eta_t &= \phi_1 \eta_{t-1} + Z_t \Rightarrow (1 - \phi_1 B) \eta_t = Z_t \\ &\Rightarrow \eta_t = (1 - \phi_1 B)^{-1} Z_t\end{aligned}$$

- ▶ Recall $\sum_{j=0}^{\infty} a^j = \frac{1}{1-a} = (1-a)^{-1}$. We have

$$\eta_t = \sum_{j=0}^{\infty} (\phi_1 B)^j Z_t = \sum_{j=0}^{\infty} \phi_1^j B^j Z_t = \sum_{j=0}^{\infty} \phi_1^j Z_{t-j}$$

$\Rightarrow$ **This is another way to show that AR(1) is a linear process**

- ▶ Here $1 - \phi_1 B$ is the **AR characteristic polynomial**, which can be used to check whether the process is **stationary** and **causal**

The Second-Order Autoregressive Process

Now consider the series satisfying

$$\eta_t = \phi_1 \eta_{t-1} + \phi_2 \eta_{t-2} + Z_t,$$

where, again, we assume that Z_t is independent of $\eta_{t-1}, \eta_{t-2}, \dots$

- ▶ The AR characteristic polynomial is

$$\phi(B) = 1 - \phi_1 B - \phi_2 B^2$$

- ▶ The corresponding **AR characteristic equation** is

$$\phi(B) = 1 - \phi_1 B - \phi_2 B^2 = 0$$

Stationarity of the AR(2) Process

- ▶ A stationary solution exists if and only if the roots of the AR characteristic equation exceed 1 in absolute value
- ▶ For the AR(2) the roots of the quadratic characteristic equation are

$$\frac{\phi_1 \pm \sqrt{\phi_1^2 + 4\phi_2}}{-2\phi_2}$$

These roots exceed 1 in absolute value if

$$\phi_1 + \phi_2 < 1, \quad \phi_2 - \phi_1 < 1, \quad \text{and } |\phi_2| < 1$$

- ▶ We say that the roots should lie outside the unit circle in the complex plane. This statement will generalize to the AR(p) case

The Autocorrelation Function for the AR(2) Process

- Yule-Walker equations:

$$\eta_t = \phi_1 \eta_{t-1} + \phi_2 \eta_{t-2} + Z_t$$

$$\eta_t \eta_{t-h} = \phi_1 \eta_{t-1} \eta_{t-h} + \phi_2 \eta_{t-2} \eta_{t-h} + Z_t \eta_{t-h}$$

$$\Rightarrow \gamma(h) = \phi_1 \gamma(h-1) + \phi_2 \gamma(h-2)$$

$$\Rightarrow \rho(h) = \phi_1 \rho(h-1) + \phi_2 \rho(h-2),$$

$$h = 1, 2, \dots$$

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$$h = 1, 2, \dots$$

- Setting $h = 1$, we have

$$\rho(1) = \phi_1 \underbrace{\rho(0)}_{=1} + \phi_2 \underbrace{\rho(-1)}_{=\rho(1)} \Rightarrow \rho(1) = \frac{\phi_1}{1-\phi_2}$$

The Autocorrelation Function for the AR(2) Process

- Yule-Walker equations:

$$\begin{aligned}\eta_t &= \phi_1 \eta_{t-1} + \phi_2 \eta_{t-2} + Z_t \\ \eta_t \eta_{t-h} &= \phi_1 \eta_{t-1} \eta_{t-h} + \phi_2 \eta_{t-2} \eta_{t-h} + Z_t \eta_{t-h} \\ \Rightarrow \gamma(h) &= \phi_1 \gamma(h-1) + \phi_2 \gamma(h-2) \\ \Rightarrow \rho(h) &= \phi_1 \rho(h-1) + \phi_2 \rho(h-2),\end{aligned}$$

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- $\rho(2) = \phi_1 \rho(1) + \phi_2 \rho(0) = \frac{\phi_2(1-\phi_2) + \phi_1^2}{1-\phi_2}$

The Variance for the AR(2) Model

Taking the variance of both sides of AR(2) equations:

$$\eta_t = \phi_1 \eta_{t-1} + \phi_2 \eta_{t-2} + Z_t,$$

yields

$$\begin{aligned}\gamma(0) &= \text{Var}(\phi_1 \eta_{t-1} + \phi_2 \eta_{t-2}) + \text{Var}(Z_t) \\ &= (\phi_1^2 + \phi_2^2)\gamma(0) + 2\phi_1\phi_2\gamma(1) + \sigma^2 \\ &= (\phi_1^2 + \phi_2^2)\gamma(0) + 2\phi_1\phi_2 \left(\frac{\phi_1\gamma(0)}{1 - \phi_2} \right) + \sigma^2 \\ &= \frac{(1 - \phi_2)\sigma^2}{(1 - \phi_2)(1 - \phi_1^2 - \phi_2^2) - 2\phi_2\phi_1^2} \\ &= \left(\frac{1 - \phi_2}{1 + \phi_2} \right) \frac{\sigma^2}{(1 - \phi_2)^2 - \phi_1^2}\end{aligned}$$

The General Autoregressive Processes

Consider the p th-order autoregressive model:

$$\eta_t = \phi_1 \eta_{t-1} + \phi_2 \eta_{t-2} + \cdots + \phi_p \eta_{t-p} + Z_t.$$

► **AR characteristic polynomial and equation:**

$$\phi(B) = 1 - \phi_1 B - \phi_2 B^2 - \cdots - \phi_p B^p, \quad \phi(B) = 0.$$

► **Yule–Walker equations:**

$$\rho(1) = \phi_1 + \phi_2 \rho(1) + \cdots + \phi_p \rho(p-1),$$

$$\rho(2) = \phi_1 \rho(1) + \phi_2 + \cdots + \phi_p \rho(p-2),$$

$$\vdots$$

$$\rho(p) = \phi_1 \rho(p-1) + \phi_2 \rho(p-2) + \cdots + \phi_p.$$

► **Variance:**

$$\gamma(0) = \phi_1 \gamma(1) + \cdots + \phi_p \gamma(p) + \sigma^2 = \frac{\sigma^2}{1 - \phi_1 \rho(1) - \cdots - \phi_p \rho(p)}.$$

Estimating the Mean of Stationary Processes

Let $\{\eta_t\}$ be **stationary** with mean μ and ACVF $\gamma(s, t) = \gamma(s - t)$

- ▶ A natural **estimator** of μ is the **sample mean**

$$\bar{\eta} = \frac{1}{T} \sum_{t=1}^T \eta_t.$$

$\bar{\eta}$ is an **unbiased** estimator of μ , i.e.

- ▶ To conduct inference for μ , we need the standard error:

$$\begin{aligned} \text{Var}(\bar{\eta}) &= \frac{1}{T^2} \text{Var} \left(\sum_{i=1}^T \eta_t \right) \\ &= \frac{1}{T^2} \sum_{s=1}^T \sum_{t=1}^T \text{Cov}(\eta_s, \eta_t) = \frac{1}{T^2} \sum_{s=1}^T \sum_{t=1}^T \gamma(s - t) \end{aligned}$$

- ▶ **Exercise:** Show

$$\text{Var}(\bar{\eta}) = \frac{1}{T} \sum_{h=-(T-1)}^{T-1} \left(1 - \frac{|h|}{T} \right) \gamma(h)$$

AR(1) Example

Suppose $\{\eta_1, \eta_2, \eta_3\}$ is an AR(1) process with $|\phi| < 1$ and **innovation** variance σ^2 . Show that the variance of $\bar{\eta}$ is $\frac{\sigma^2}{9(1-\phi^2)}(3 + 4\phi + 2\phi^2)$

Solution:

The Sampling Distribution of $\bar{\eta}$

Let

$$v_T = \sum_{h=-(T-1)}^{(T-1)} \left(1 - \frac{|h|}{T}\right) \gamma(h)$$

► If $\{\eta_t\}$ is **Gaussian** we have

$$\sqrt{T}(\bar{\eta} - \mu) \sim N(0, v_T)$$

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- ▶ The result above is **approximate** for many **non-Gaussian** time series
- ▶ In practice, we also need to **estimate** $\gamma(h)$ from the data. We will return to this later

Confidence Intervals for μ

- If $\gamma(h) \rightarrow 0$ as $h \rightarrow \infty$ then

$$v = \lim_{T \rightarrow \infty} v_T = \sum_{h=-\infty}^{\infty} \gamma(h) \text{ exists.}$$

- Further, if $\{\eta_t\}$ is **Gaussian** and

$$\sum_{h=-\infty}^{\infty} |\gamma(h)| < \infty,$$

then an **approximate large-sample** 95% CI for μ is given by

$$\left[\bar{\eta} - 1.96\sqrt{\frac{v}{T}}, \bar{\eta} + 1.96\sqrt{\frac{v}{T}} \right]$$

Strategies for Estimating $v = \sum_{h=-\infty}^{\infty} \gamma(h)$

► Parametric:

- Assume a parametric model $\gamma_{\theta}(\cdot)$, and calculate

$$\hat{v} = \sum_{h=-\infty}^{\infty} \gamma_{\hat{\theta}}(h)$$

based on the ACVF for that model

- The standard error depends on the parameters θ of the parametric model

► Nonparametric:

- Estimate v by

$$\hat{v} = \sum_{h=-\infty}^{\infty} \hat{\gamma}(h),$$

where $\hat{\gamma}(\cdot)$ is a nonparametric estimate of ACVF

Examples of Parametric Forms for v

- ▶ **i.i.d. Gaussian Noise:** $v = \gamma(0) = \sigma^2 \Rightarrow$ CI reduces to the classical case:

$$\left[\bar{\eta} - 1.96 \sqrt{\frac{\sigma^2}{T}}, \bar{\eta} + 1.96 \sqrt{\frac{\sigma^2}{T}} \right]$$

- ▶ **MA(1) process:** We have

$$\begin{aligned} v &= \sum_{h=-\infty}^{\infty} \gamma(h) = \gamma(-1) + \gamma(0) + \gamma(1) \\ &= \gamma(0) + 2\gamma(1) \\ &= \sigma^2(1 + \theta^2 + 2\theta) = \sigma^2(1 + \theta)^2 \end{aligned}$$

- ▶ **Exercise:** Show for an **AR(1)** process we have

$$v = \frac{\sigma^2}{(1 - \phi)^2}$$

An Estimator of $\gamma(\cdot)$

Goal: Estimate the autocovariance function (ACVF)

$$\gamma(h) = \text{Cov}(\eta_t, \eta_{t+h}) = \mathbb{E}[(\eta_t - \mu)(\eta_{t+h} - \mu)],$$

using data $\{\eta_t\}_{t=1}^T$.

- For $|h| < T$, define the **sample ACVF**:

$$\hat{\gamma}(h) = \frac{1}{T} \sum_{t=1}^{T-|h|} (\eta_t - \bar{\eta})(\eta_{t+|h|} - \bar{\eta}).$$

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- ▶ $\hat{\gamma}(h)$ is a **biased** estimator of $\gamma(h)$, but it is the **standard** estimator used in practice
- ▶ The collection $\{\hat{\gamma}(h)\}$ is **symmetric** (i.e., $\hat{\gamma}(h) = \hat{\gamma}(-h)$, $\forall h$)

An Estimator of $\gamma(\cdot)$

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- **Properties:**

- $\hat{\gamma}(h)$ is a **biased** estimator of $\gamma(h)$, but it is the **standard** estimator used in practice
- The collection $\{\hat{\gamma}(h)\}$ is **symmetric** (i.e., $\hat{\gamma}(h) = \hat{\gamma}(-h)$, $\forall h$)
- It is **non-negative definite**; that is, for all vectors $\mathbf{a}$, $\mathbf{a}^T \Sigma \mathbf{a} \geq 0$, where Σ is the Toeplitz covariance matrix whose (i, j) -th element is $\gamma(i - j)$

The Sample Autocorrelation Function

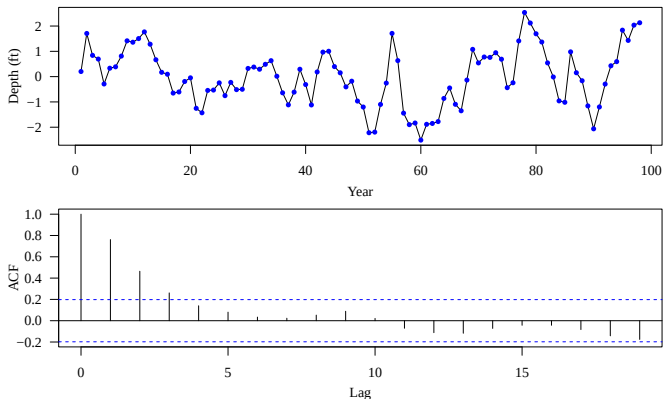
- ▶ The **sample autocorrelation function** (ACF) is defined for $|h| < T$ by

$$\hat{\rho}(h) = \frac{\hat{\gamma}(h)}{\hat{\gamma}(0)}.$$

- ▶ **Rule of thumb:** Box and Jenkins (1976) recommend using $\hat{\rho}(h)$ and $\hat{\gamma}(h)$ only for $\frac{|h|}{T} \leq \frac{1}{4}$ and $T \geq 50$
- ▶ This is because estimates $\hat{\rho}(h)$ and $\hat{\gamma}(h)$ are unstable for large $|h|$ as there will be no enough data points going into the estimator

Calculating the Sample ACF in R

- ▶ We use `acf` function to calculate the sample ACF
- ▶ Lake Huron Example



Asymptotic Distribution of the Sample ACF [Bartlett, 1946]

Let $\{\eta_t\}$ be a stationary process we suppose that the ACF

$$\boldsymbol{\rho} = (\rho(1), \rho(2), \dots, \rho(k))^T$$

is estimated by

$$\hat{\boldsymbol{\rho}} = (\hat{\rho}(1), \hat{\rho}(2), \dots, \hat{\rho}(k))^T$$

► For large T

$$\hat{\boldsymbol{\rho}} \sim N_k(\boldsymbol{\rho}, \frac{1}{T}W),$$

where N_k is the k -variate normal distribution and W is an $k \times k$ covariance matrix with (i, j) element defined by

$$w_{ij} = \sum_{h=1}^{\infty} a_{ih}a_{jh}, \quad 1 \leq i \leq k, \quad 1 \leq j \leq k$$

where $a_{ih} = \rho(h+i) + \rho(h-i) - 2\rho(h)\rho(i)$

Using the ACF as a Test for White Noise

For a white noise process with finite variance, for each fixed $h \neq 0$,

$$\hat{\rho}(h) \sim N\left(0, \frac{1}{T}\right).$$

This suggests a **diagnostic** for white noise:

1. Plot the sample ACF $\hat{\rho}(h)$ against lag h .
2. Add approximate 95% bounds at $\pm \frac{1.96}{\sqrt{T}}$ (blue dashed lines in R).
3. For white noise, individual sample autocorrelations should generally lie within these bounds, with occasional exceedances expected by chance.

The Portmanteau Test [Box and Pierce, 1970] for White Noise

Suppose we wish to test:

$H_0 : \{\eta_1, \eta_2, \dots, \eta_T\}$ is an white noise sequence

$H_1 : H_0$ is false

► Under H_0 ,

$$\hat{\rho}(h) \sim N(0, \frac{1}{T}) \stackrel{d}{=} \frac{1}{\sqrt{T}} N(0, 1)$$

► Hence

$$Q = T \sum_{i=1}^k \hat{\rho}^2(h) \sim \chi_{df=k}^2$$

► We **reject** H_0 if $Q > \chi_k^2(1 - \alpha)$, the $1 - \alpha$ quantile of the chi-squared distribution with k degrees of freedom

Ljung-Box Test [Ljung and Box, 1978]

Ljung and Box [1978] showed that

$$Q_{LB} = T(T+2) \sum_{h=1}^k \frac{\hat{\rho}^2(h)}{T-h} \sim \chi_k^2.$$

The Ljung-Box test can be more powerful than the Portmanteau test

Both the Portmanteau Test (aka Box-Pierce test) and Ljung-Box test can be carried out in R using the function `Box.test()`

New Weighted Portmanteau Tests [Fisher & Gallagher, 2012]

$$Q_W = T \sum_{h=1}^k w_h \hat{\rho}(h)^2, \quad w_h = \frac{k+1-h}{k} \quad (\text{linearly decreasing}).$$

- ▶ **Why:** Better finite-sample behavior and power than Box-Pierce/Ljung-Box, especially for larger k or underfit ARMA
- ▶ **Null:** $Q_W \rightsquigarrow$ weighted sum of χ_1^2 ; use a Gamma approximation for p -values
- ▶ **Nonlinearity:** Apply to squared residuals to detect ARCH/GARCH
- ▶ **R:** `WeightedPortTest::Weighted.Box.test()`